

Can five cells build a spaceship

A Game of Life discovery worksheet

Explorer name _____

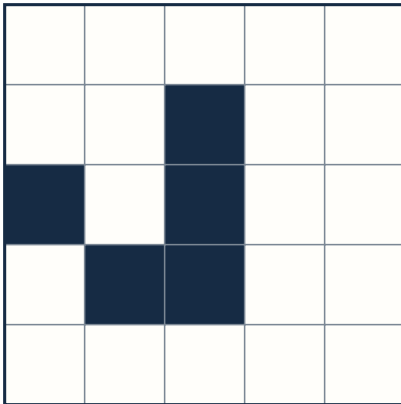
Date _____

The story

Maya shades five cells in a tiny universe. She follows the rules of Life. After a few turns, the same shape appears in a new place. No one pushed it. The rules made it travel. That kind of travelling pattern is called a spaceship. This small spaceship is a glider.

Meet the glider

Generation 0



Five live cells

The rules

- A live cell survives with 2 or 3 live neighbors.
- An empty cell is born with exactly 3 live neighbors.
- Every other cell is empty in the next generation.

A useful definition

A spaceship is a pattern that returns to the same shape after a fixed number of turns, but in a different place.

THINK 1

Before you calculate anything: what do you predict this glider will do after 4 turns? Circle one.

☐ disappear ☐ stand still ☐ return somewhere else ☐ grow forever

THINK 2

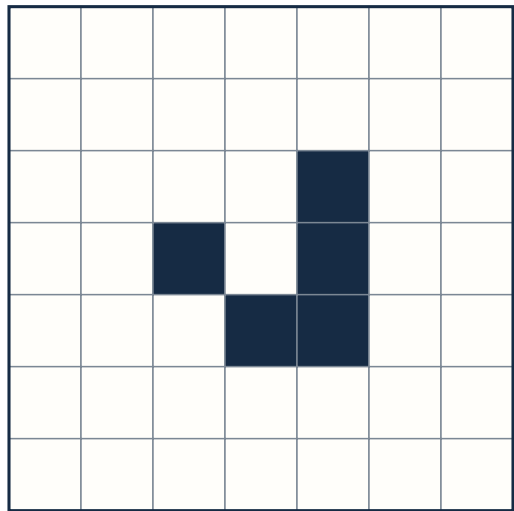
What evidence would convince you that this is really a spaceship?

Track one turn at a time

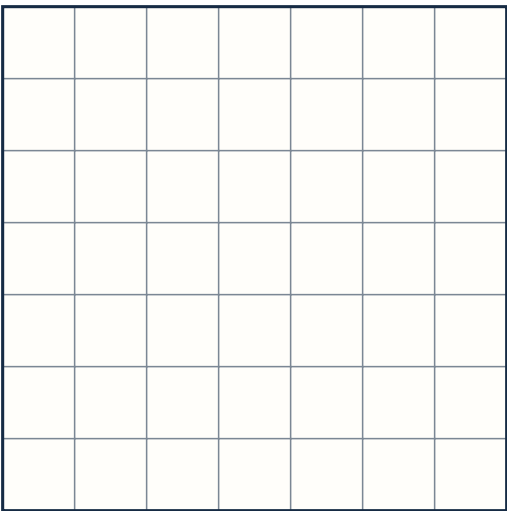
Every cell checks the old generation before the next one appears

The first grid is Maya's launch pattern. Fill the next two grids by applying the rules to every cell at the same time. Do not let a cell use a neighbor from the new grid.

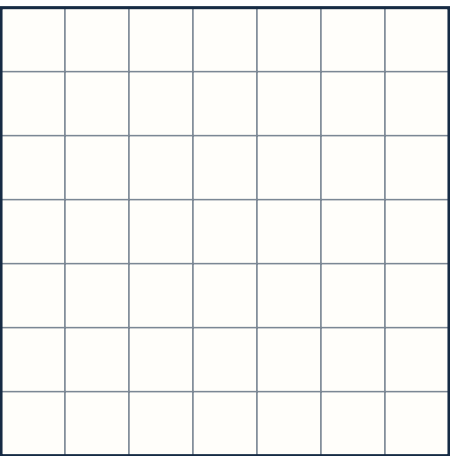
Generation 0 given



Generation 1 you calculate



Generation 2 you calculate



Flight log

Live cells in Generation 0

Live cells in Generation 1

Live cells in Generation 2

My best observation

Great work. On the next page, calculate two more turns and decide whether the glider really is a spaceship.

Finish the flight

Now look for a repeated shape in a new location

Continue the same experiment. Calculate Generations 3 and 4. Then compare Generation 4 with the launch pattern on page 2.

Generation 3

Generation 4

Spaceship detective

THINK 3

Is Generation 4 the same shape as Generation 0? What makes you say so?

THINK 4

If it moved, describe its move. How many squares did it shift sideways and up or down?

THINK 5

A spaceship repeats after a number of turns called its period. What is the glider's period?

Bonus for a pattern hunter

Could a pattern repeat its shape but stay in the same place? Draw or describe what you think that would be called.

Invent a Life machine

Try to make a pattern with a personality

Choose a challenge. Shade cells in a grid, then calculate at least one next generation somewhere else. Your pattern does not need to work on the first try. Good mathematicians keep evidence.

Still life

does not change

Oscillator

repeats in one place

Spaceship

repeats somewhere else

Your original experiment

Generation 0

Lab notebook

THINK 6

What are you trying to build?

THINK 7

Your prediction before testing

THINK 8

What actually happened?

One more question to chase

Can you invent a pattern with more than one kind of behavior?